Design Calculation Sheet - GridStore-D3

VBF-T2R1-CALC-01 | t2_r1 grid energy storage | 2026-08-25 | values mechanically sourced from parameter set / simulation outputs

1. Input parameters

Parameter	Base (Chen2020)	Final (GridStore-D3)	Source
Nominal cell capacity [A.h]	5.0	5.0	Chen2020 dump
Positive electrode thickness [m]	7.56e-05	7.56e-05	Chen2020 dump
Negative electrode thickness [m]	8.52e-05	8.52e-05	Chen2020 dump
Positive electrode porosity	0.335	0.4	r6_d3_params.json
Negative electrode porosity	0.25	0.4	r6_d3_params.json
Positive particle radius [m]	5.22e-06	2.5e-06	r6_d3_params.json
Negative particle radius [m]	5.86e-06	2e-06	r6_d3_params.json
Electrolyte conductivity [S.m-1]	Nyman2008 f(T)	3.0	r6_d3_params.json (constant override)
Electrolyte diffusivity [m2.s-1]	Nyman2008 f(T)	2.5e-09	r6_d3_params.json (constant override)
Cation transference number	0.2594	0.35	r6_d3_params.json
SEI kinetic rate constant [m.s-1]	1e-12	1e-16	r6_d3_params.json (coating bridge)
Voltage cut-offs [V]	2.5 / 4.2	2.5 / 4.2	Chen2020 dump

2. Capacity and energy

Item	Formula	Value	Source
1C discharge capacity [Ah]	run-pyamm 1C_discharge	5.066256277024658	cell/r6_d3_1c.json:capacity_ah
Discharge energy [Wh]	trapezoid integral of voltage_v x I_1C over time_s; I_1C = 5.0 A	18.4542	cell/r6_d3_energy.json:energy_wh

3. Energy density

Item	Formula	Value	Source
Gravimetric ED [Wh/kg]	energy / mass (layers only, electrolyte excluded)	465.62164250933915	cell/r6_d3_energy.json:energy_density_wh_kg
Volumetric ED [Wh/L]	energy / volume (layer stack x area)	894.8725767864692	cell/r6_d3_energy.json:energy_density_wh_l
Criterion check	465.62 >= 327.18	PASS	entry-0 criteria.stage2.energy_density_wh_kg

4. N/P and mass

Item	Formula	Value	Source
N/P ratio	(33133x0.9014x85.2um)/(63104x0.2700x75.6um)	1.9755	Chen2020 dump; initial-Li-content caliber (set defines initial concentrations, not x0/x100 windows)
Positive electrode	75.6um x 0.60 x 3262	147.96 g/m2	r6_d3_energy.json:layer_kg_m2
Negative electrode	85.2um x 0.60 x 1657	84.71 g/m2	idem
Positive current collector	16um x 2700	43.20 g/m2	idem
Negative current collector	12um x 8960	107.52 g/m2	idem
Separator	12um x 0.53 x 397	2.52 g/m2	idem
TOTAL mass	sum x 0.1027 m2	39.633 g	r6_d3_energy.json:mass_kg

5. Process parameters

Parameter	Formula	Value	Unit	Source
Positive areal density	75.6um x (1-0.40) x 3262	147.96	g/m2	calc-energy layer_kg_m2
Negative areal density	85.2um x (1-0.40) x 1657	84.71	g/m2	calc-energy layer_kg_m2
Positive compaction density	3262 x (1-0.40) / 1000	1.957	g/cm3	design-spec process formula (divide by 1000)
Negative compaction density	1657 x (1-0.40) / 1000	0.994	g/cm3	design-spec process formula
Electrolyte fill amount	pore volume 7.185 cm3 x 1.2 g/cm3 x 100%	8.62	g	electrolyte density 1.2 g/cm3 = literature value (annotated)
Formation recommendation	0.1C CC to 4.2V, 25C, 2 cycles	-		design-recommended; production value requires tuning